Hardware Security of DNNs: Motivation and background

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Need of improving DNN security

- Until recently, DNNs were employed only
  - For simple applications, such as image classification,
  - used on-site computing/manufacturing resources.
  - used high-end systems equipped with ECC
- Now, DNNs are used for mission critical applications
- DNN training is outsourced to clouds and DNN-accelerator design to a global supply chain.
- AI market will reach $118.6 billion by 2025
- It takes huge time and resources to get trained hyper-parameters of a DNN, so better to steal them!

Types/examples of attacks

- By leaking DNN architecture, adversary can design a better DNN before you!
- Attacker can change specific weights to degrade accuracy or cause malfunction
- Targeted misclassification attack: change prediction of only specific images so that the average classification accuracy is not affected.
  - make the DNN infer that a flu-patient has COVID
- Feeding “imperceptibly different” images to cause misclassification
- In radiography and sonography images, by leaking the foreground entity’s shape, attacker can infer a subject’s body-organ and health-condition

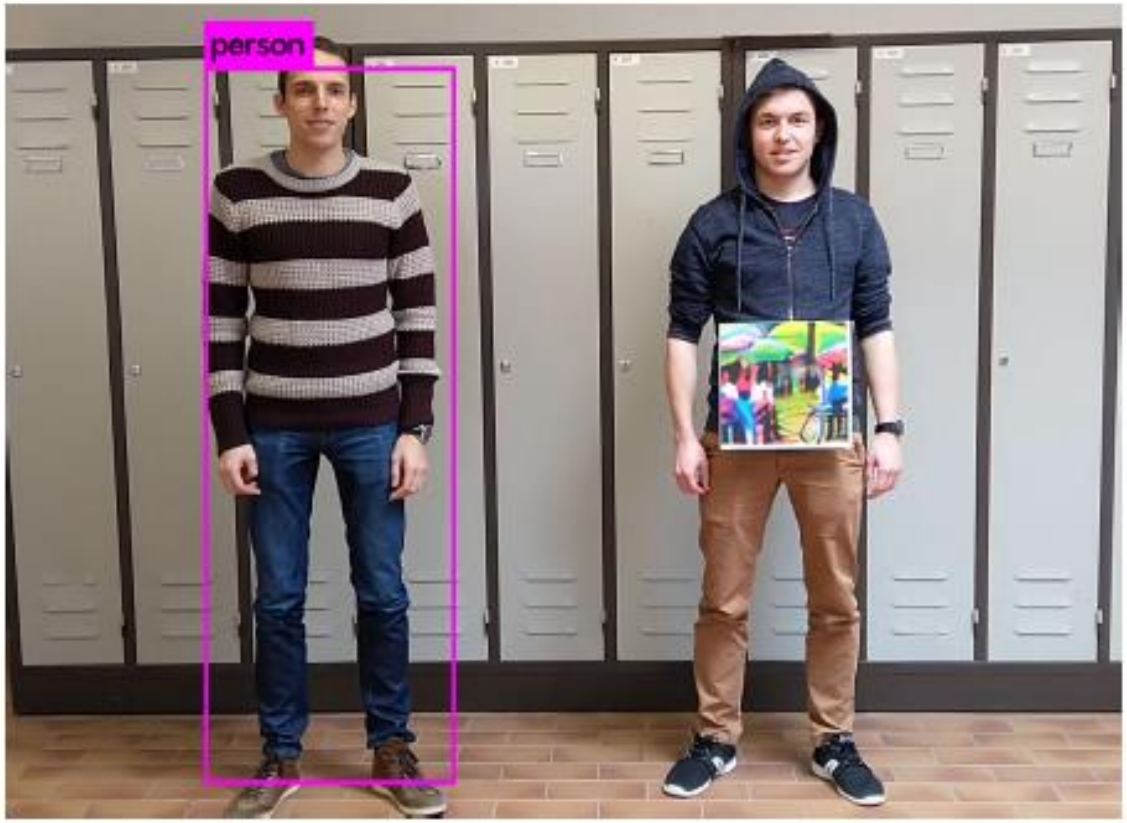
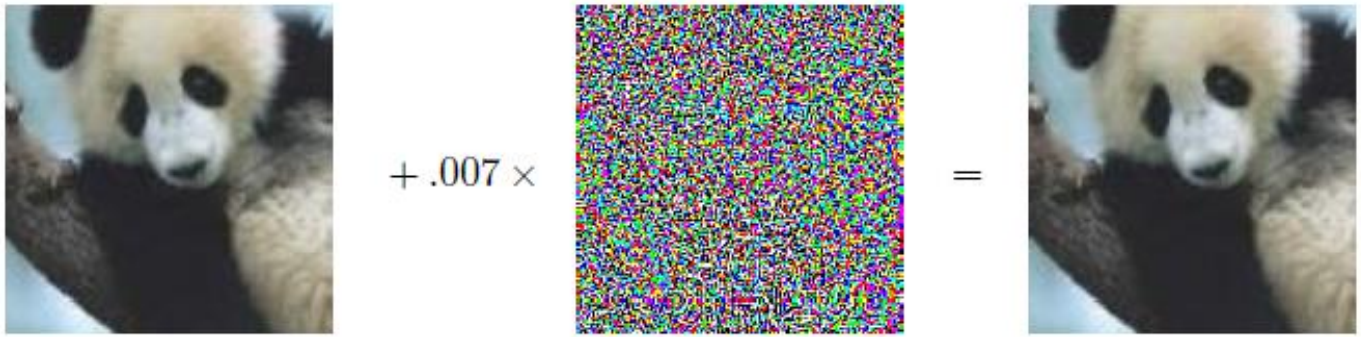


Figure 1: We create an adversarial patch that is successfully able to hide persons from a person detector. Left: The person without a patch is successfully detected. Right: The person holding the patch is ignored.



x

"panda"
57.7% confidence

$+ .007 \times$

$\text{sign}(\nabla_{\boldsymbol{x}} J(\boldsymbol{\theta}, x, y))$

"nematode"
8.2% confidence

$=$

$x + \epsilon\text{sign}(\nabla_{\boldsymbol{x}} J(\boldsymbol{\theta}, x, y))$

"gibbon"
99.3 % confidence

Goodfellow et al., Explaining and harnessing adversarial examples

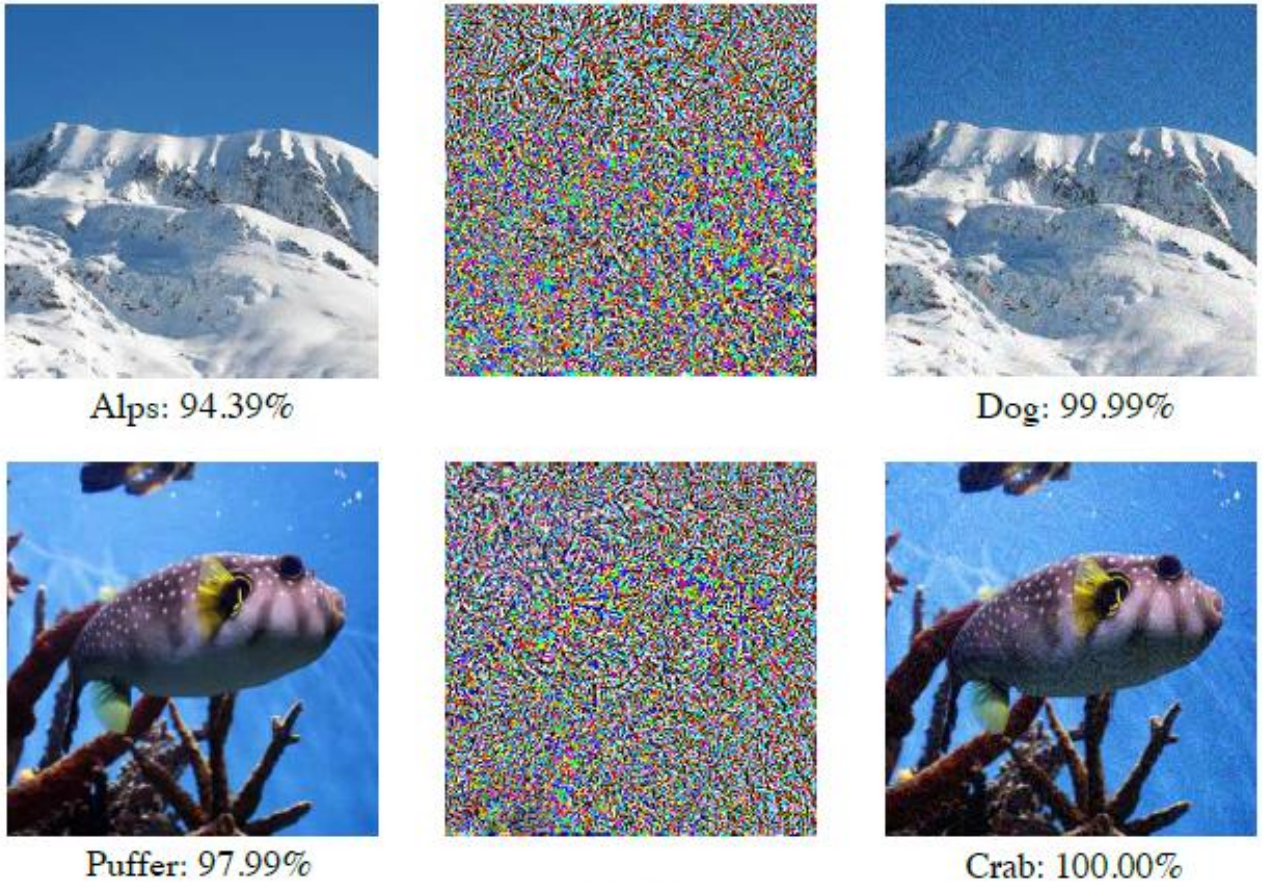


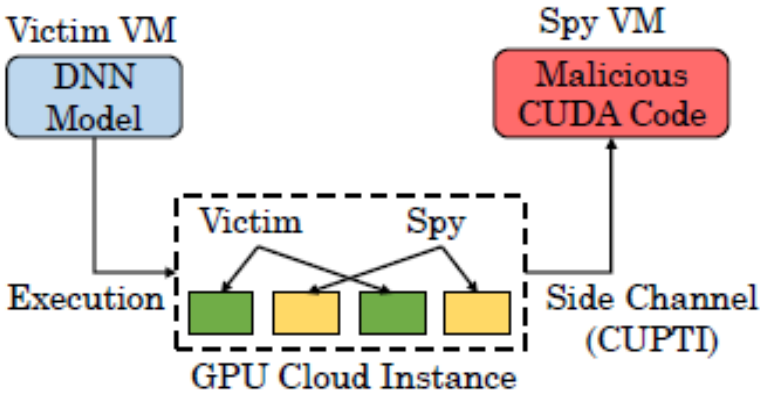
Figure 1. We show two adversarial examples generated by the proposed momentum iterative fast gradient sign method (MI-FGSM) for the Inception v3 [22] model. **Left column**: the original images. **Middle column**: the adversarial noises by applying MI-FGSM for 10 iterations. **Right column**: the generated adversarial images. We also show the predicted labels and probabilities of these images given by the Inception v3.

Why attacking DNNs is difficult

- DNNs have insulation to fault-injection attacks due to over-parameterization, presence of batch-norm, dropout, ReLU.
- Distinguishing kernels of different layers is difficult.
- Finding layer boundaries is difficult.
  - If a CONV operation is merged with activation-function and pooling, they cannot be distinguished because off-chip access is made only once for these layers.

Side channel attack

- Here, spy uses a side-channel to infer what the victim is doing
  - Example: Visitor determining if a person is in the office by overhearing outside the office



- Cache side-channel attack: In multicore processors with shared last-level cache, a spy can know about its co-running victim application from its cache behavior